Nathan Zimmerman

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WORK HISTORY

Software Engineer II, Nelnet 10/2023 – Present

- Built core features within a greenfield Payment Plans & Billing domain for a multi-tenant SaaS platform serving educational institutions, spanning C#/.NET microservices, Angular, MongoDB, and Redis.
- Designed payment plan offering and enrollment workflows across admin and payer BFF apps, supporting plan setup, payment rules, deterministic UTC runtime status derivation, and automated payment rollbacks.
- Developed a shared Angular/npm payment schedule calculator with a stateless BFF admin preview, generating installment scenarios from rules around down payments, cadence, fees, and completion dates.
- Integrated payment, banking, disclosure, pricing, and messaging services, using MongoDB multi-document transactions, a transactional outbox, and API key roles to provision multi-tenant RBAC permissions at signup.
- Built two-step installment charging and event-driven returned-payment handling that applied tenant return policies including installment rescheduling, fee assessment, grace-period holds, and plan cancellation.

Software Development Engineer, Amazon 05/2022 – 04/2023

- Engineered risk-guidance applications using Java, Scala, Spring, and AWS to screen third-party sellers against denied-party lists and financial crime risk signals, including money-laundering indicators.
- Remediated critical security issues by enabling encryption for SQS/SNS/S3 via KMS, tightening IAM policies to least privilege, and upgrading vulnerable dependencies.
- Executed EC2 host migrations across IAD/PDX Availability Zones for 10+ applications and 100+ environments, validating cutovers with alarms and dashboards to maintain zero downtime.
- Reduced operational support load by resolving pipeline failures, monitoring alarms, and deployment errors across production application workflows.

Software Developer, Radian Group Inc. 09/2019 – 04/2022

- Contributed to the launch and maintenance of MyRadian, a React and Node.js application suite for valuation and asset-management services, supporting feature delivery, reviews, bug fixes, and production support.
- Troubleshot production issues and improved performance using AWS CloudWatch, X-Ray, DynamoDB, Splunk, and SQL across modern and legacy C#/.NET and Java applications.
- Automated database reconciliation workflows with Python scripts, eliminating recurring manual operations and improving data accuracy.

PROJECTS

- Wheelhouse - Built a family calendar for a wall touch screen and phones with ASP.NET Core, Angular and PostgreSQL. Syncs with Google Calendar and pushes live updates over SignalR. | calendar.nathanzimmerman.com
- Rabbit Ears - Built a 90s-style TV that plays folders of old shows as channels, using .NET, Angular, SignalR and WebGL CRT shaders, with a phone as the remote. | rabbitears.nathanzimmerman.com
- Know Ball - Built daily sports trivia games, mostly NBA, with an ASP.NET Core API, Angular and PostgreSQL. A nightly import loads every NBA player since 1947. | knowball.nathanzimmerman.com
- Brick Breaker Resume - Built a Phaser game that turns a Word résumé into a playable level, where every word is a brick, with responsive scaling and saved high scores. | resume.nathanzimmerman.com

SKILLS

C#, .NET, Azure, AWS, REST APIs, Microservices, MongoDB, SQL, Redis, TypeScript, Angular, React, Java, CI/CD

EDUCATION

AS, Computer Science, Delaware County Community College, 2018

BA, Psychology, (History minor), Rutgers University School of Arts and Sciences, 2016